CSE-411
Computer Networks

Comer, Chapter 8:
LAN Technology and Network Topologies

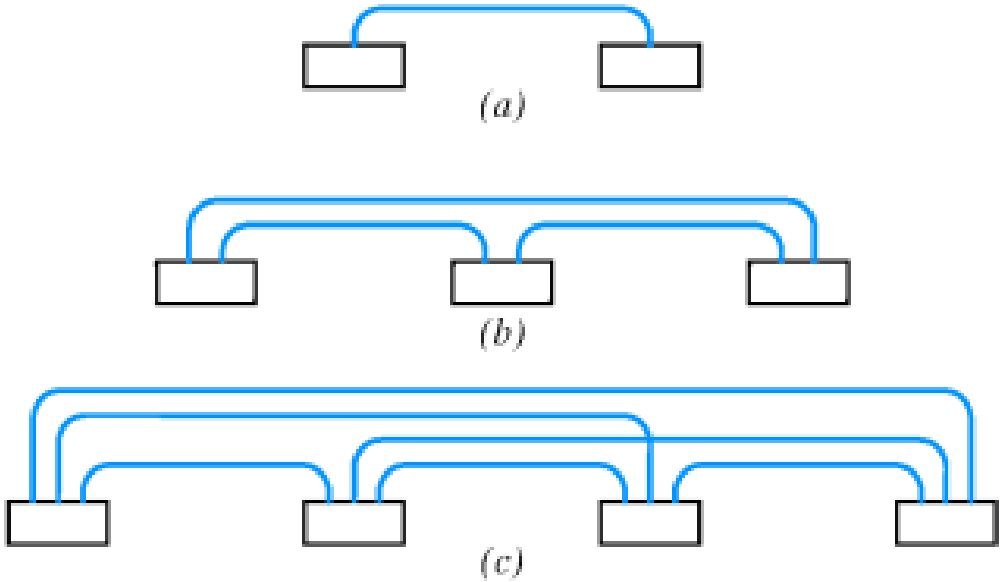
Direct Point to Point Communication

Direct Point to Point Communication

- Computers connected by communication channels that each connect exactly two computers
- Forms *mesh* or *point-to-point* network
- Allows flexibility in communication hardware, packet formats, etc.
- Provides security and privacy because communication channel is not shared

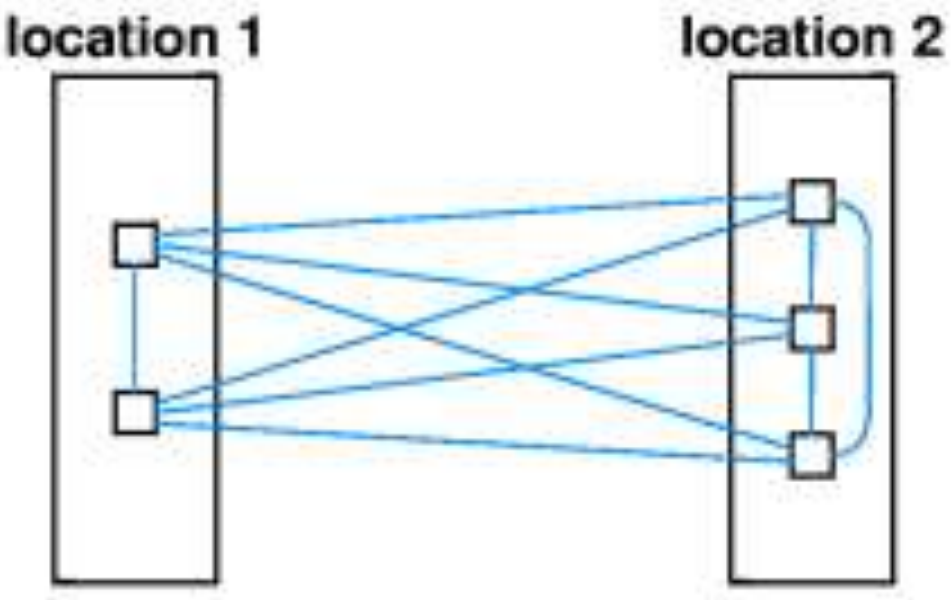
Connections in a Point-to-point Network

- The number of wires grows as square of number of computers
- For n computers, connections = $(n^2 - n) / 2$.



Connections in a Point-to-point Network

- Connections between buildings can be prohibitive.
- Adding a new computer requires n-1 connections.



Reducing the number of communication channels

- LANs developed in late 1960s and early 1970s
- Key idea - reduce number of connections by *sharing* connections among many computers
  - Computers take turns
  - TDM Must include techniques for synchronizing use

Growth of LAN Technologies

- LAN technologies reduce cost by reducing number of connections
- **But** … attached computers compete for use of shared connection
- Local communication almost exclusively LAN
- Long distance almost exclusively point-to-point
  - SMDS
  - ATM

Locality of Reference

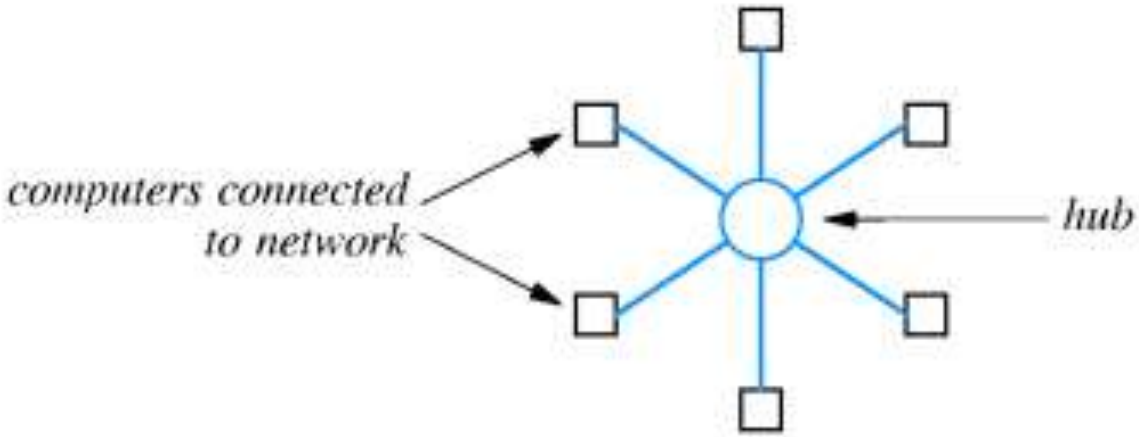
- Principle of *locality of reference* helps predict computer communication patterns:
  - *Spatial* (or *physical*) locality of reference - computers likely to communicate with other computers that are located nearby
  - *Temporal* locality of reference - computers are likely to communicate with the same computers repeatedly
- Thus LANs are effective because of spatial locality of reference, and temporal locality of reference may give insight into which computers should be on a LAN

LAN Topologies

- Networks may be classified by shape
- Three most popular:
  - Star
  - Ring
  - Bus

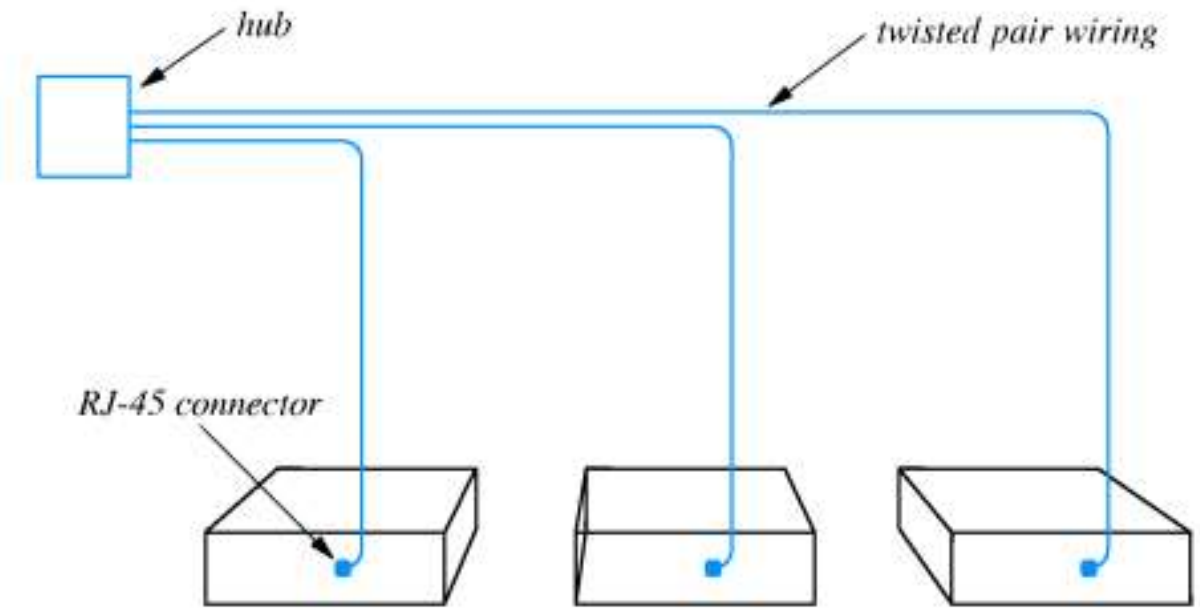
Star Topology

- All computers attach to a single central point.
- The central point is sometimes called a *hub*.



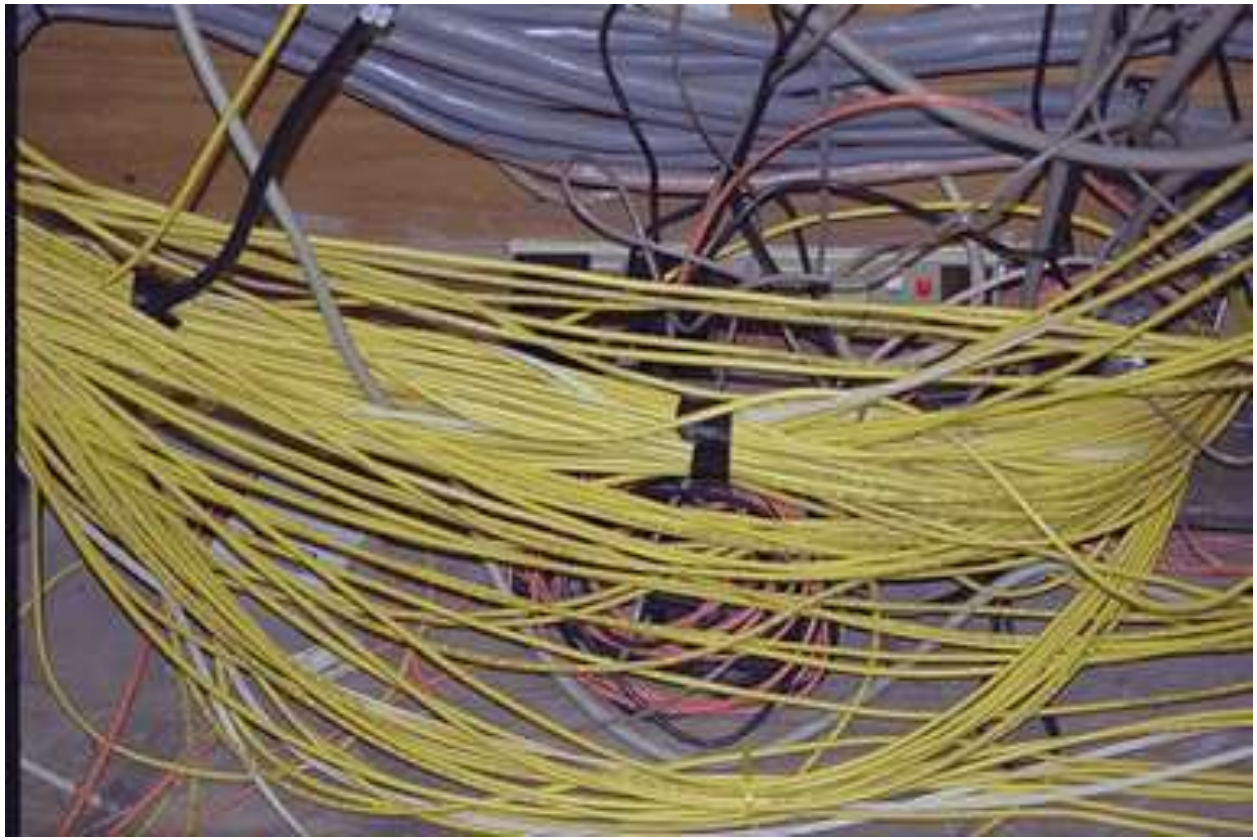
Star Topology in Practice

- Previous diagram is idealized; usually, connecting cables run in parallel to computers:



Star Topology in Practice

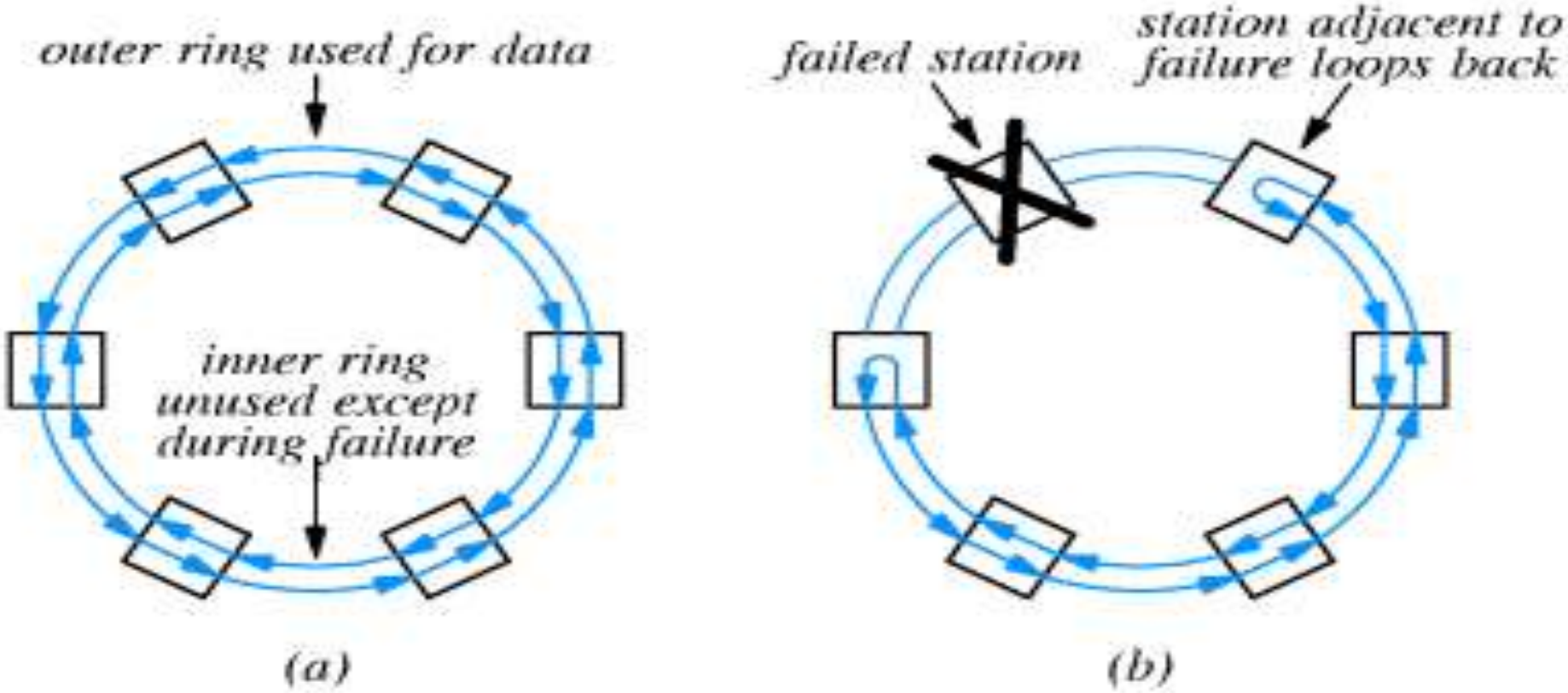
- The result often looks like this:



Ring Topology

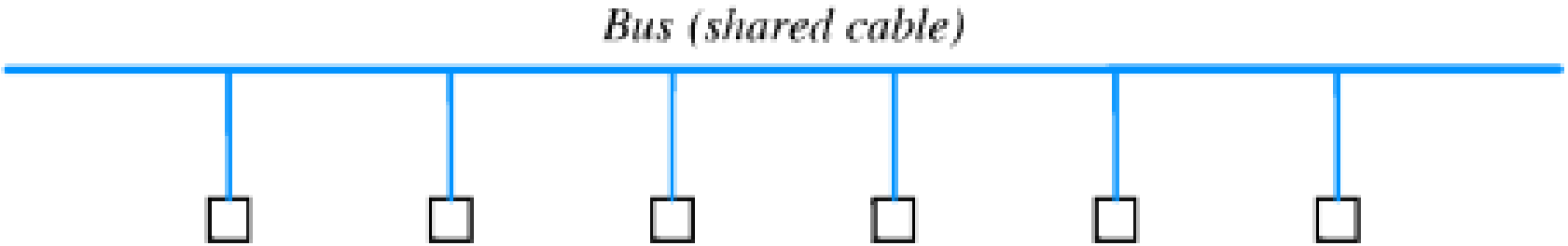
- Computers connected in a closed loop
- First passes data to second, second passes data to third, and so on
- In practice, there is a short connector cable from the computer to the ring
- Ring connections may run past offices with connector cable to socket in the office:

Ring Topology



Bus Topology

- Single cable connects all computers.
- Each computer has connector to shared cable.
- Computers must synchronize and allow only one computer to transmit at a time.



Why Have Multiple Topologies?

- Each topology has advantages and disadvantages:
  - Ring: eases synchronization; may be disabled if any cable is cut.
  - Star: easier to manage and more robust; requires more cables.
  - Bus: requires fewer cables; may be disable if cable is cut.
- UNO has used all three topologies; star topology is most widely used in PKI.
- Industry is settling on star topology as most widely used

Ethernet

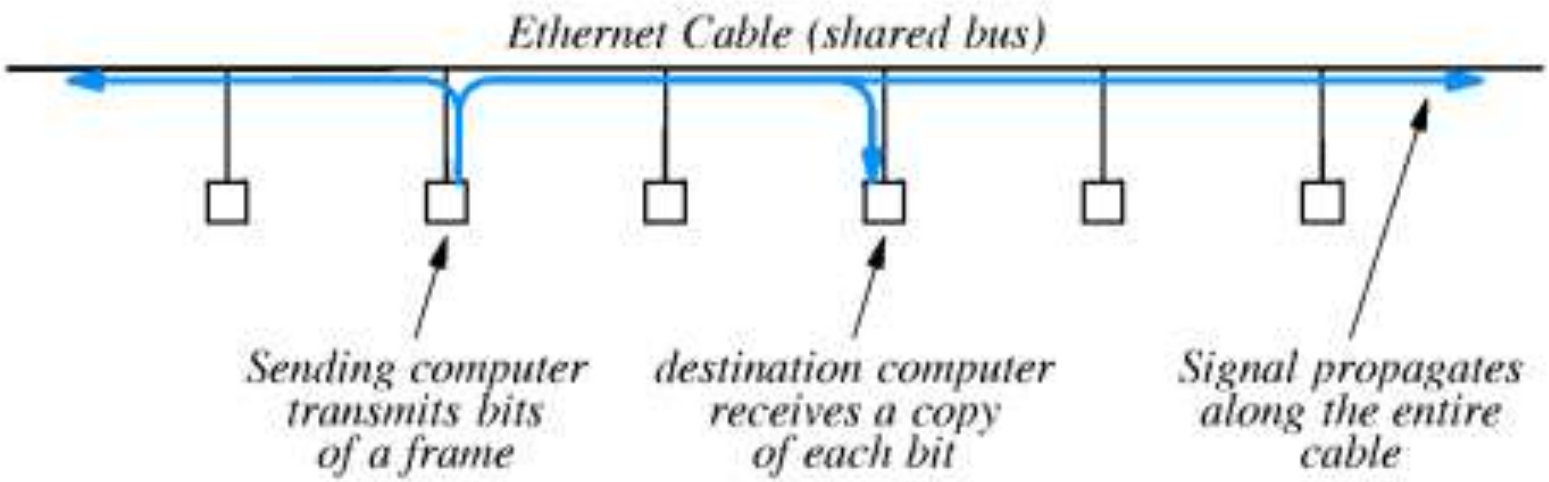
- Widely used LAN technology
  - Invented at Xerox PARC (Palo Alto Research Center) in 1970s
  - Defined in a standard by Xerox, Intel and Digital - *DIX* standard
  - Standard now managed by IEEE - defines formats, voltages, cable lengths, ...
- Uses bus topology
  - Single coax cable - the *ether*
  - Multiple computers connect
- One Ethernet cable is sometimes called a *segment*
  - Limited to 500 meters in length
  - Minimum separation between connections is 3 meters

Ethernet Speeds

- Originally Ethernet ran at 3Mbps.
- The previous standard speed was 10Mbps.
- Most new installations now use *Fast Ethernet*, which operates at 100Mbps.
- Ethernet is also now commonly used at higher speeds (e.g. 1000Mbps) on backbone cables.

Ethernet Operation

- One computer transmits at a time.
- The signal is a modulated carrier which propagates from the transmitter in both directions along the length of the segment.



CSMA

- There is no central control managing when computers transmit on ether.
- Ethernet employs CSMA to coordinate transmission among multiple attached computers.
- *Carrier Sense with Multiple Access*
  - Multiple access - multiple computers are attached and any of these can be a transmitter.
  - Carrier sense – the computer wanting to transmit tests ether for a carrier before transmitting.

Collision Detection

- Even with CSMA, two computers may transmit simultaneously.
  - Both check the ether at same time, find it idle, and begin transmitting.
  - The window for transmission depends on the speed of propagation in the ether.
- Signals from two computers will interfere with each other if they are present at the same place at the same time.
- Overlapping frames is called a *collision .*
  - Collisions cause no harm to hardware.
  - Data from both frames is garbled.

Ethernet Collision Detection

- Ethernet interfaces include hardware to detect transmissions:
  - They monitor the outgoing signal (listen while sending)
  - A garbled signal is interpreted as a collision.
- After a collision is detected, the computer immediately stops transmitting.
- So, Ethernet uses CSMA/CD to coordinate transmissions.

Recovery from a Collision

- The computer that detects a collision sends a special signal to force all other interfaces to detect collision.
- That computer then waits for ether to be idle before transmitting again
  - If both computers involved in the collision wait same length of time, the frames will collide again.
  - The standard specifies the maximum delay, and both computers choose a random delay less than the maximum.
- After waiting, the computers use carrier sense to avoid subsequent collisions.
  - The computer with the shorter delay will go first.
  - But other computers (not involved in the collision) may transmit first.

Exponential Backoff

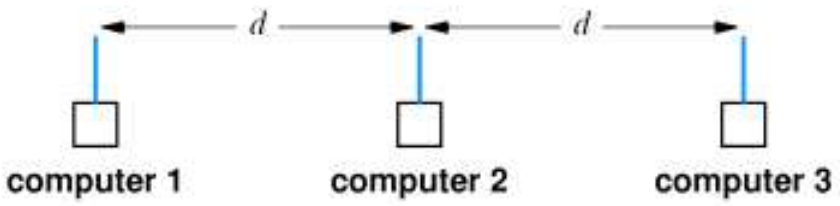
- Even with random delays, collisions may occur.
- This is especially likely with busy network segments.
- Computers double their delay period with each subsequent collision.
- This reduces the likelihood of a sequence of collisions.

Wireless LANs

- These typically use radio signals at 900MHz.
- The most frequently used data rate is 2Mbps (faster is more difficult to achieve reliably).
- The shared medium is radio instead of coaxial cable.

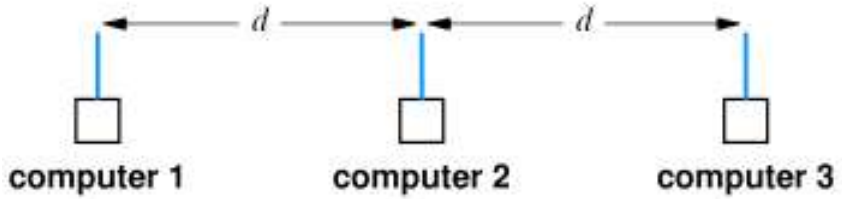
Limited Wireless Connectivity

- In contrast with wired LAN, not all participants may be able to reach each other, because of
  - low signal strength
  - propagation blocked by walls, etc.
- We can't depend on CD; not all participants may hear the signal.



CSMA/CA

- Wireless uses *collision avoidance* rather than collision detection
  - Transmitting computer sends very short message to receiver
  - Receiver responds with short message reserving slot for transmitter
- Response from receiver is *broadcast* so all potential transmitters receive reservation



Collisions

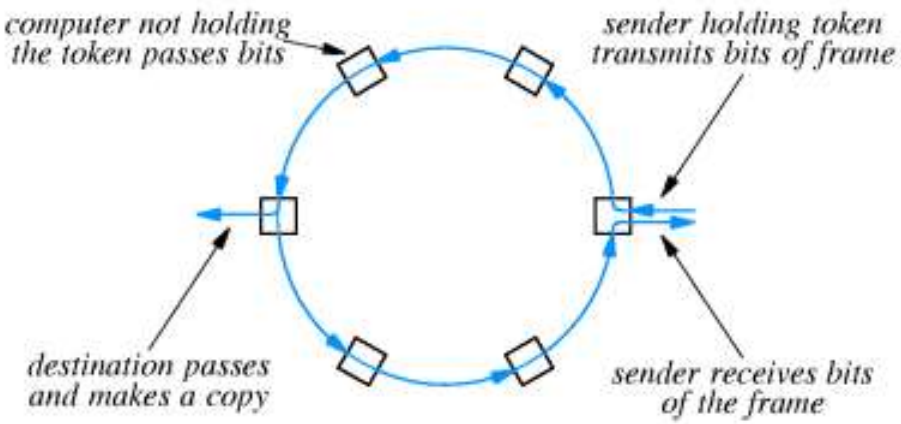
- Receiver may receive simultaneous requests.
  - This results in a collision at receiver.
  - **Both** requests are lost.
  - Neither transmitter receives reservation; both use backoff and retry.
- Receiver may receive closely spaced requests
  - It selects one of these.
  - The selected transmitter sends message.
  - The transmitter not selected uses backoff and retries.

LocalTalk

- This is a LAN technology that uses bus topology.
- It is the interface included with all Macintosh computers (at least until recently).
- It operates at a relatively low speed - 230.4Kbps.
- It is low cost ("free" with a Macintosh); easy to install and connect.
- It uses CSMA/CD.

Token Ring

- Many LAN technologies that use ring topology use *token passing* for synchronized access to the ring
- The ring itself is treated as a single, shared communication medium
- Bits pass from transmitter, past other computers and are copied by destination
- Hardware must be designed to pass token even if attached computer is powered down



Transmission Around a Token Ring

- As in other LANs, only one computer can successfully transmit at a time.
- With token ring, this is further enforced by requiring the transmitting computer to possess a *token*, a small message giving the right to transmit to the token holder.
- The token can be held only for a specified period of time.

Using the Token

- When a computer wants to transmit, it waits for the *token*.
- After transmission, computer transmits token on ring to the next computer.
- The next computer ready to transmit receives the token and then transmits.

The Token and Synchronization

- Because there is only one token, only one computer will transmit at a time
  - The token is a short, reserved frame that cannot appear in data.
  - The hardware must regenerate the token if it is lost.
- The token gives a computer permission to send one frame.
  - If all computers are ready to transmit, this enforces “round-robin” access to the shared media
  - If no computers are ready to transmit, the token circulates around the ring.

IBM Token Ring

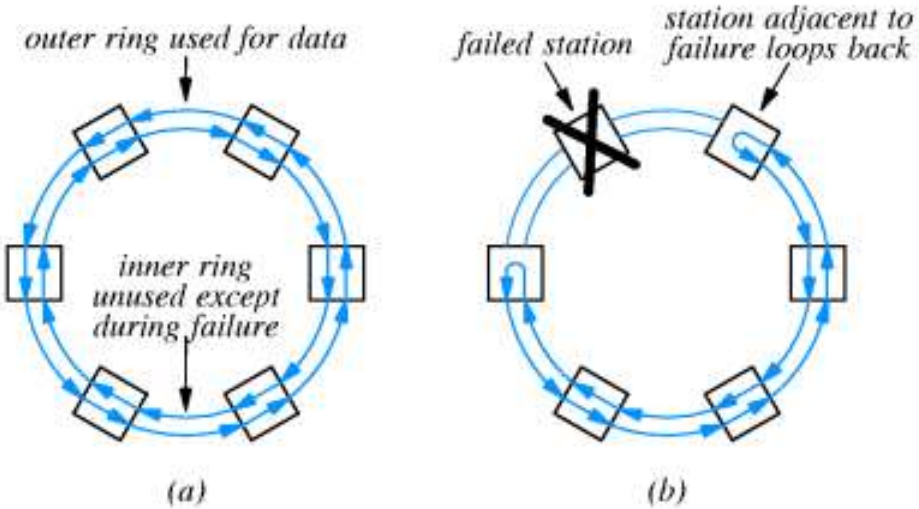
- Widely used (second to Ethernet).
- Originally 4Mbps, now 16Mbps.
- Uses a special connector cable between the computer and ring interface.

FDDI

- *Fiber Distributed Data Interconnect* (FDDI) is another ring technology
  - Uses fiber optics between stations
  - Transmits data at 100Mbps
- Uses pairs of fibers to form two concentric rings

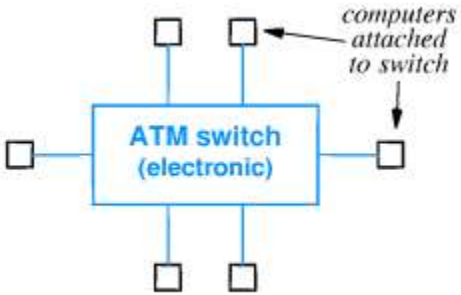
FDDI and Reliability

- FDDI uses *counter-rotating* rings in which data flows in opposite directions.
- In case of fiber or station failure, the remaining stations *loop back* and reroute data through spare ring.
- All stations automatically configure the loop back by monitoring the data ring.



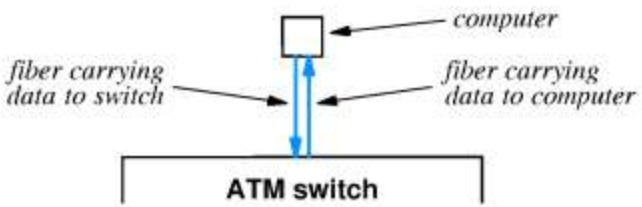
ATM - A Star Network

- *Asynchronous Transfer Mode* technology consists of electronic packet switches to which computers can connect.
- ATM switches form a *hub* into which computers connect in a *star* topology.
- Computers get point-to-point connections - data from a transmitter is routed directly through the hub switches to t



ATM Details

- ATM transmits data at over 100 Mbps.
- Fiber optic cables are used to connect each computer to the switch.
- Each connection requires two fibers.

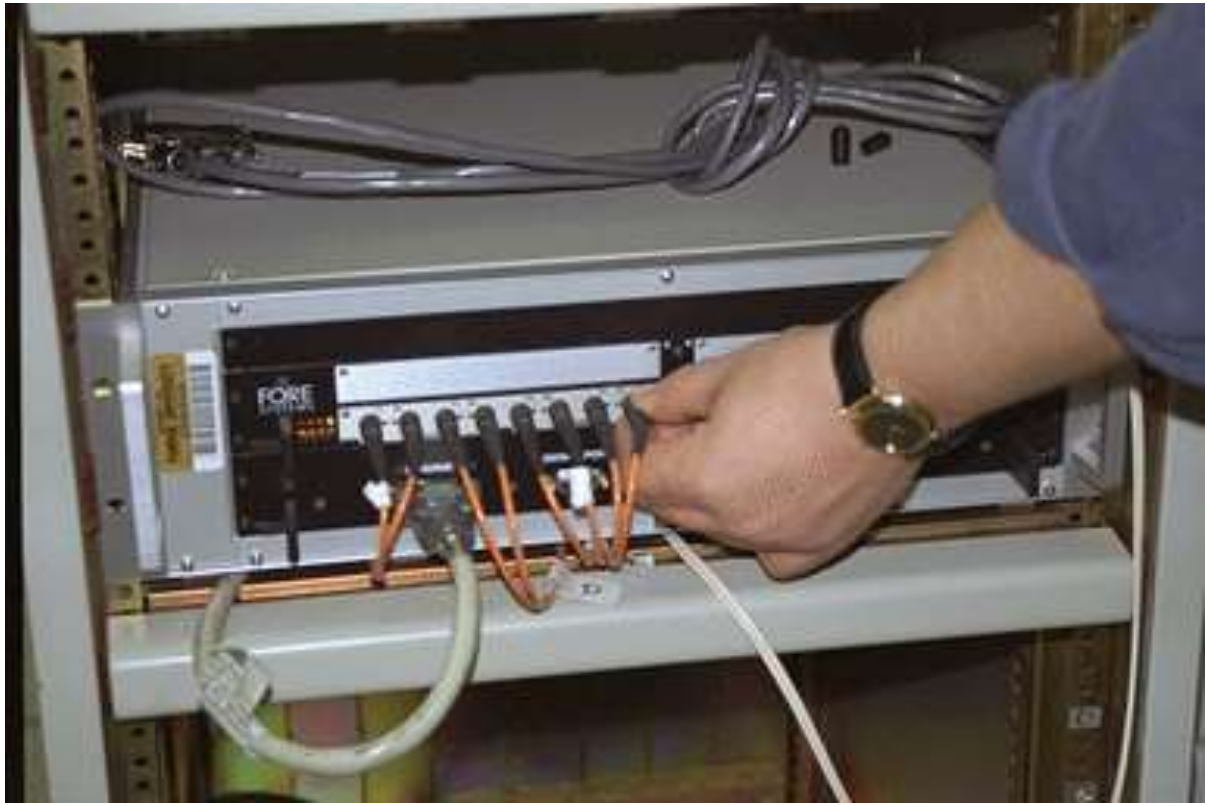


ATM Illustrations - 1

- The orange fiber cables connect to the computer:

ATM Illustrations - 2

- An ATM switch; again note the orange fiber cables.



Summary

- LAN technologies use shared communication media to interconnect multiple computers over short distances
- Transmitting computer has exclusive use of communication medium; computers must synchronize transmission and share available capacity
- LAN topologies:
  - Star
  - Ring
  - Bus